Q1

```
[asjadraza@Asjads-MacBook-Air-2 OS % ./Q1
Producer 0 produced: 0 at position 0
Consumer 1 consumed: 100 from position 0
Consumer 0 consumed: 100 from position 0
Producer 1 produced: 100 at position 0
Producer 0 produced: 0 at position 2
Producer 1 produced: 100 at position 2
Consumer 1 consumed: 100 from position 2
Consumer 0 consumed: 100 from position 2
Producer 1 produced: 100 at position 4
Producer 0 produced: 0 at position 0
Consumer 1 consumed: 100 from position 4
Consumer 0 consumed: 100 from position 4
Done!
asjadraza@Asjads-MacBook-Air-2 OS %
```

Q2

Each fork is protected by a semaphore initialized to 1, so only one philosopher can acquire it at a time. Any apparent overlap in output is due to unsynchronized print statements, not an actual race condition.

```
[asjadraza@Asjads-MacBook-Air-2 OS % ./Q2
Philosopher 0 is thinking...
Philosopher 0 is hungry, waiting for forks...
Philosopher 0 picked up fork 0
Philosopher 0 picked up fork 1, now EATING!
Philosopher 0 put down forks, done eating
Philosopher 2 is thinking...
Philosopher 2 is hungry, waiting for forks...
Philosopher 2 picked up fork 2
Philosopher 2 picked up fork 3, now EATING!
Philosopher 2 put down forks, done eating
Philosopher 3 is thinking...
Philosopher 3 is hungry, waiting for forks...
Philosopher 3 picked up fork 3
Philosopher 3 picked up fork 4, now EATING!
Philosopher 3 put down forks, done eating
Philosopher 4 is thinking...
Philosopher 4 is hungry, waiting for forks...
Philosopher 4 picked up fork 0
Philosopher 4 picked up fork 4, now EATING!
Philosopher 4 put down forks, done eating
Philosopher 1 is thinking...
Philosopher 1 is hungry, waiting for forks...
Philosopher 1 picked up fork 1
Philosopher 1 picked up fork 2, now EATING!
Philosopher 1 put down forks, done eating
All philosophers are done!
asjadraza@Asjads-MacBook-Air-2 OS %
```